Teacher-in-the-Loop LLM Exercise Authoring for Cambridge KET and PET

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PROBLEM

Supply falls short.

Learners practise weekly for two to three years, and the official past papers run out well before that.

Authoring by hand is slow.

A teacher must hold the part's structure, the CEFR band and defensible distractors at once; harder item types get skipped.

Chat tools ignore the exam.

They write fluent text that misses the part's structure and vocabulary list, and they keep no record of the teacher's corrections.

SOLUTION AND METHOD

Each draft is checked before review

A model drafts the part the teacher chose. Deterministic program checks test the schema, handbook counts, vocabulary, key spread and the key against the passage; a model-based check answers each cloze gap without the key. A failed blocking check starts an automatic retry.

Every correction is recorded

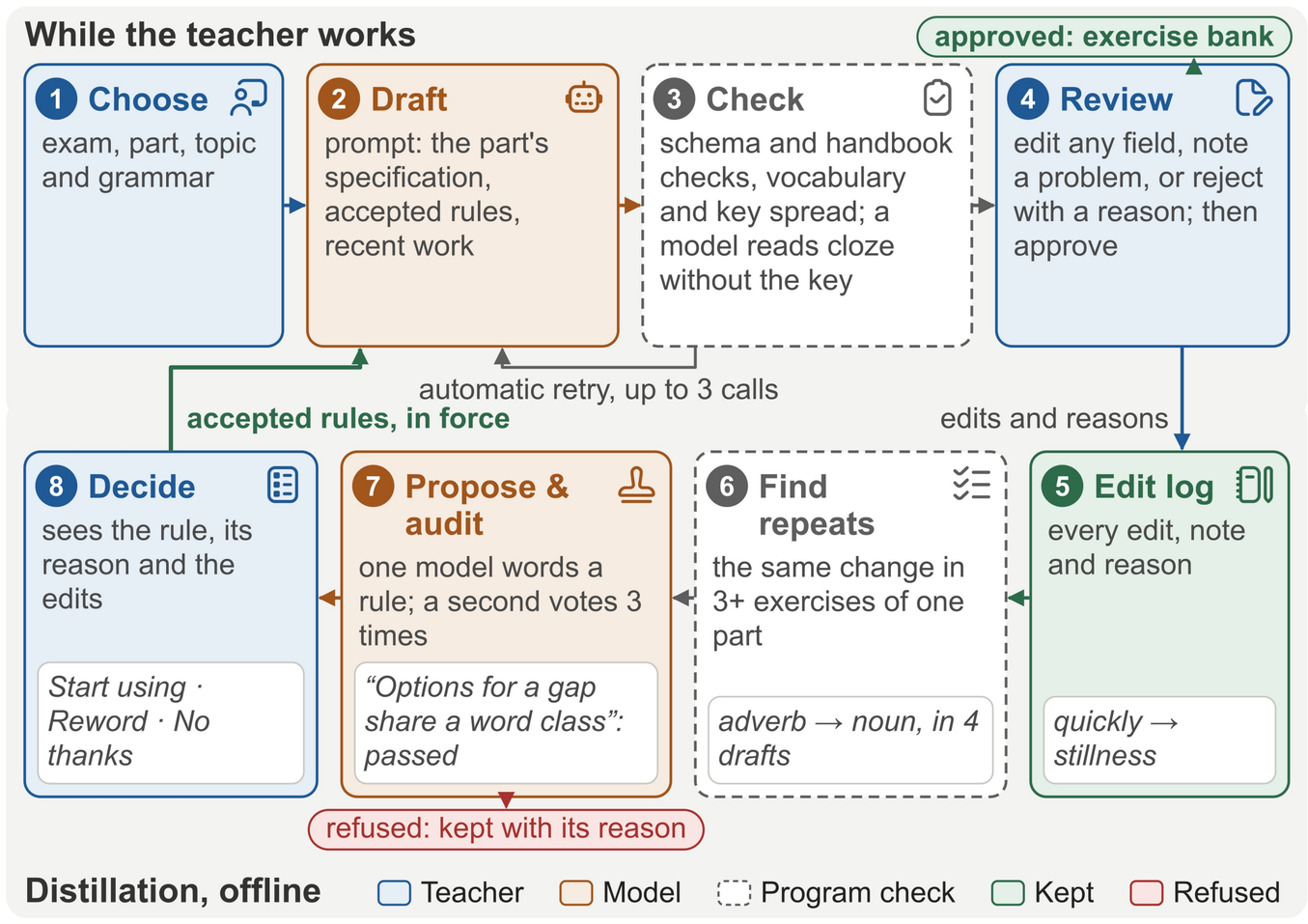
The untouched draft is kept beside the teacher's version.

Repeated corrections become proposals

Offline, a program counts edits with the same exam part and field shape (option A of any question); three or more exercises make a candidate. One model words a rule and a reviewer model votes three times. A majority only places the proposal before the teacher.

Only accepted rules reach the prompt

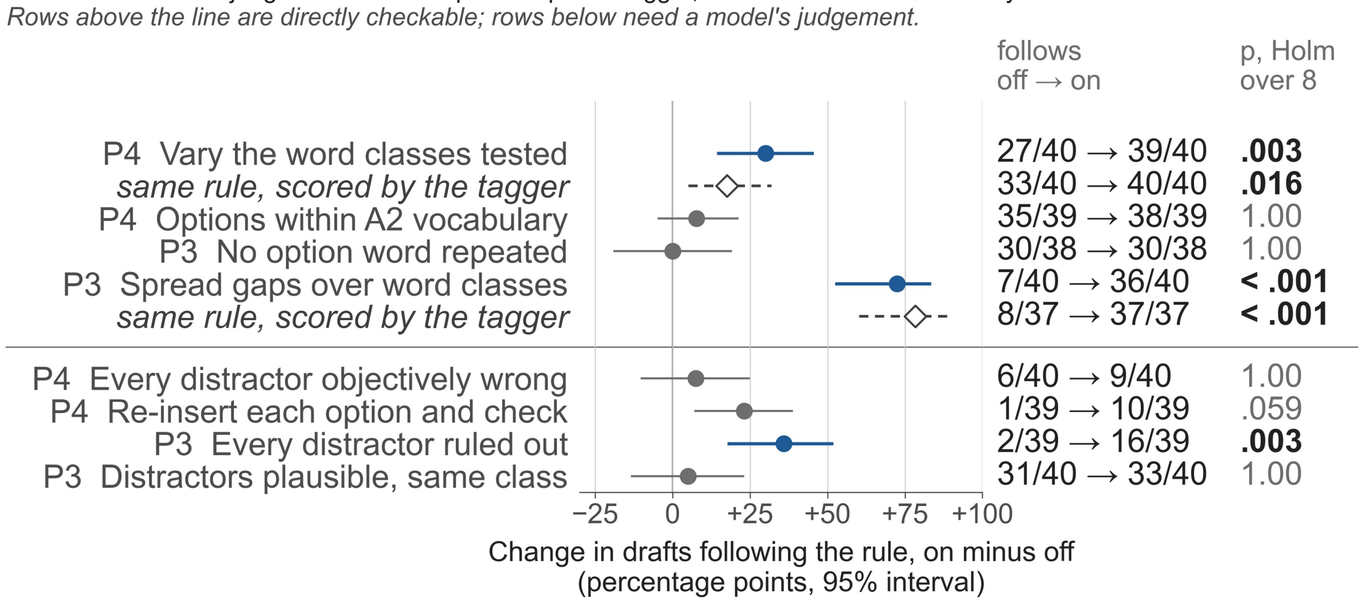
The teacher accepts, rewords or refuses each proposal and can stop any rule later.



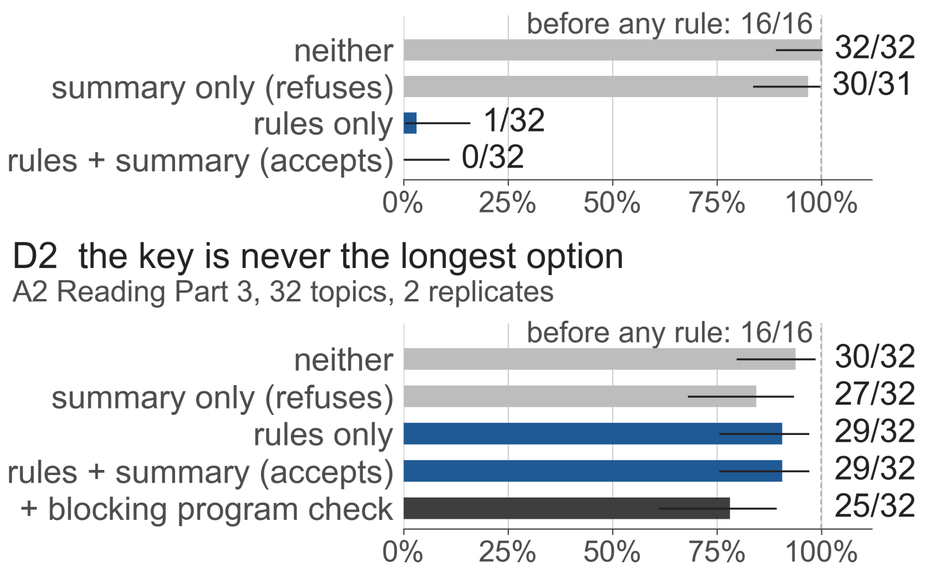
RESULTS

Of **612** logged generations with a model call, **488 (79.7%)** passed on the first call and **122** needed an automatic retry. The first failed check was the key against the passage in **51**, the cloze check in **44**, the handbook in **19** and the schema in **11**. **54** regression scripts test the shipped code.

filled marker: model judge diamond: part-of-speech tagger, labels counted automatically
Rows above the line are directly checkable; rows below need a model's judgement.



Scored by model judges, not teacher data.
Each rule measured with the teacher's four accepted rules on.
p: Holm over the eight judge tests; the tagger rows are unadjusted.



Synthetic, offline, not teacher data. D2 prompt rule p = .69. Blocking check vs prompt rule, same topics (30/32): p = .18.

Teacher study. Two teachers in three sessions accepted all eight proposed rules; one wrote that they had no direct say in how observations became rules. Teachers can now reword and rank rules and switch off the recent-work summary.

Prompt rule or program check. As a prompt instruction, "the key is never the longest option" left 31 of 84 questions with the longest option as key, against 32 with no rule. A blocking deterministic program check cut it to 17 (p = .012) at 2.81 against 1.06 model calls per draft. For synthetic teachers, drafts needing correction fell from 30/32 to 25/32 (p = .18).

Model-based screening. Against the author's blind labels, the model that looks for a second correct answer flags too often (sensitivity 0.89, specificity 0.61). A flag changes no item; the teacher decides.

https://fyp.z-chen.dev